

CHE 205 - Computational Methods in Chemical Engineering - Fall 2025

Instructor: Shafigh Mehraeen, Associate Professor

Office: Department of Chemical Engineering, Engineering Innovation Building (EIB) 242
929 W. Taylor Street, Chicago, IL 60607

Office Phone: (312) 996-8734

Email: tranzabi@uic.edu

Office Hours: In-person, T: 10:45-11:45am, virtual, W: 3:30pm-4:30pm, available by e-mail at other times

TA: Zac Adesanya, tadesa5@uic.edu

TA's Office Hours: TTh, 12:30-1:30pm, EIB 176

Class Times and Location: TTh, 8:00-9:15am, EIB 124

Grading Policy:

1. Homework 35%, two midterms (9/25, 11/6, during class time), each 20%, and one final exam (Thu, 12/11, 8:00-10:00am), 25%.
2. Any evidence of copying will be addressed in accordance with the university's academic integrity policies.
3. Assignments are due at 11:59pm on the due date, and must be submitted online through blackboard. Solutions will be posted 24 hours after the due date. Late assignment (delivered before solutions are posted) will be penalized 50% on the grade of that assignment. Assignments turned in after solutions are posted will not be accepted.
4. If there is only one late assignment throughout the semester, late penalty will be lifted from that assignment (i.e. 50% of the graded credit for that assignment will be returned).
5. Regrading is only considered if it is asked for within one week of the date the assignment is handed to the students.

Required Text: Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB, by V.J. Law. ISBN: 978-1-4665-7534-9, CRC Press, 2013.

- Ch. 1: Roots of a single nonlinear equation

- Ch. 2: Visual Basic for applications programming
- Ch. 3: Linear algebra and systems of linear equations
- Ch. 4: Numerical differentiation and integration
- Ch. 5: Ordinary differential equations (initial value problems)
- Ch. 6: Ordinary differential equations (boundary value problems)
- Ch. 8: Partial differential equations
- Ch. 7: Regression analysis and parameter estimation
- Ch. 9: Linear programming, nonlinear programming, nonlinear equations, and nonlinear regression using Solver

Supplemental Materials: If you need to learn more about VBA programming, you may find many tutorials on the internet. You can also consult with the following websites:

- http://excelvbatutor.com/vba_tutorial.html
- http://www.homeandlearn.org/basic_mathematical_operators.html
- <http://www.automateexcel.com/vba-code-examples>
- 5 videos, posted on blackboard, all about syntax in VBA programs

Objectives: By taking this course, you will be able to

1. write VBA script for most algorithms in chemical engineering.
2. solve many mathematical problems in thermodynamics and transport numerically using spreadsheets.

Prerequisite(s): Credit or concurrent registration in CHE 201, and MATH 210

Attendance Policy

1. If you miss a class, you're responsible for the materials covered in that class.
2. There will be random bonus in-class exercises for which you can get a bonus credit only if you attend the class, otherwise you will not get any credit for the bonus in-class exercises.
3. If there is a handout in addition to the course textbook, it will be posted on the blackboard.
4. Exam attendance is mandatory. You do not get credit for any exam you miss. Makeup exams will only be given in the case of an emergency, which requires an official letter of accommodation from UIC, and will be decided in a case-by-case basis.